CS 501 E1 · PROJECT UPDATE 1 · NOVEMBER 10, 2026

Habitable

A heat-and-repair record for Boston renters: our working vertical slice, from sensor to landlord notice to server.

TEAM

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SCRUM MASTER THIS WEEK

Ankita Patra

THE VERTICAL SLICE

One flow, end to end

1
Sensor
Simulated now, Bluetooth next

2
Room
Insert-only readings, one transaction each

3
Rules
68 / 64 °F, Sept 15 to May 31

4
Chart
Step limit plus Open-Meteo weather

5
Issue
Opens itself after 2 hours cold

6
Notice
Dated email letter to the landlord

7
Sync
Readings and issues to our API

Every step is real code. Nothing on screen is hard-coded.

Compose UI → ViewModel → repository → Room, Open-Meteo and our Express API.

ABOUT FIVE MINUTES, ON THE EMULATOR

Live demo

Priyanshu drives

1. Set up the apartment (validated, saved)
2. Start monitoring: foreground service
3. Rotate: state survives
4. Replay a cold night: 97 readings
5. Go offline: weather from cache

Ankita drives

1. Home turns red: 6.6° below the limit
2. Open the issue that opened itself
3. Send notice: email to the landlord
4. Request inspection: timeline grows
5. Sync now: readings and issue on the server

ARCHITECTURE

Room is the source of truth

Compose UI
7 destinations · type-safe Navigation Compose · stateless *Content composables

ViewModels
One StateFlow<UiState> per screen · combine + stateIn · survives rotation

Repositories
Readings · Weather · Issues · Apartment · Sync: the only code that touches data

Room + DataStore
7 tables · settings

Open-Meteo
Retrofit, cached

Our API
Express + JWT

Domain, pure Kotlin

No Android imports, unit-tested at the legal boundaries

- HeatRules
- ViolationDetector
- DataQualityChecker
- WeatherCrossCheck
- NoticeTemplate

Sensor: foreground service → SensorSource (simulated or BLE) → ReadingsRepository

PERSISTENCE AND API EVIDENCE

Local data, a third-party API and our own

Room + DataStore

- 7 tables; readings are insert-only
- Flags, violations and auto-issue in one transaction
- Settings and sign-in token in DataStore
- Shown live in Database Inspector

Open-Meteo

- Hourly outdoor temperature via Retrofit
- Cached in Room, so it works offline
- Loading, error and stale states with Retry
- Cross-check: is the sensor outdoors?

Our Express API

- JWT sign-in, bcrypt passwords
- POST /readings, deduped by sensor and time
- PUT /issues, upserted by client id
- PostgreSQL, or in-memory for demos

76
automated tests, all passing

57
unit: rules, detector, letters, sync

6
on-device: Room pipeline, Compose UI

13
backend API

WORK STATUS

Done, in progress, next

DONE · UPDATE 1

- Setup with validation, saved
- Simulated sensor: live and replay
- Chart vs. step limit and weather
- Automatic heat issue, escalation
- Notice letter by email
- Readings and issue sync
- Redesigned UI

IN PROGRESS

- Bluetooth thermometer on real hardware (blocked: device ordered)
- Backend deploy to Render + Neon
- Notices stored as server records

NEXT · UPDATE 2, NOV 24

- Photos hashed at capture
- PDF evidence packet
- Reminders after certification
- Address geocoding
- Accessibility pass

BACKLOG, ROLES AND SCOPE

Prioritised backlog

ID	Story	Owner	Status
US-3	Chart vs. legal limit	Priyanshu	Done
US-5	Issues + automatic heat issue	Ankita	Done
US-6	Notice + reminders	Ankita	Partial
US-1	Account + apartment	Both	Done
US-10	Open-Meteo weather	Priyanshu	Done
US-2	BLE thermometer + photo	Priyanshu	Blocked
US-4	Photos hashed at capture	Ankita	Update 2
US-7	PDF evidence packet	Ankita	Update 2
US-9	Backend deployed	Priyanshu	Partial

SCRUM MASTER

Ankita, week of Nov 10

Rotates weekly. Runs the board, Tuesday planning and Friday review.

SCOPE CHANGES

- Canvas chart instead of Vico (sign-off?)
- One cold night = one issue
- Notice sent from the tenant’s email app
- Geocoding moved to sprint 2

Nothing added or dropped.

RISKS AND BLOCKERS

What could go wrong, and our plan

Type	Risk	Mitigation
Device: blocker	Bluetooth thermometer not yet in hand	Decoder unit-tested; simulator uses the same pipeline
Device	Android stops Bluetooth sampling in the background	Foreground service; gaps are flagged, never hidden
Permissions	Notifications or Bluetooth denied	Asked only when needed; Bluetooth scan with neverForLocation
Privacy	Address, photos, landlord details are sensitive	Stay on the phone; only readings and an issue summary sync, after sign-in
API	Open-Meteo or our laptop backend is down	Weather cached with Retry; backend moves to Render + Neon
Usability	Legal text read as legal advice	“Documentation only, not legal advice” on screens and letters
Schedule	Camera and PDF are both large	Camera in sprint 2, PDF in sprint 3; stretch work frozen

REPOSITORY EVIDENCE

How we work

Issues and project board

Every backlog story is a GitHub Issue with an owner, size and milestone, on a To do / In progress / Done board.

Branches and pull requests

feat/evidence-issue-sync (Ankita) and docs/update1 (Priyanshu), each reviewed by the other. main is protected.

Continuous integration

GitHub Actions on every push: backend tests, Android build and unit tests.

Cadence and disclosure

Daily check-in; Tuesday planning, Friday review. AI use disclosed in docs/AI_USE.md.

> Lesson learned: from Oct 20 to 24 we committed straight to main from one machine. Since Nov 8 every change goes through a reviewed pull request.

PLAN TO UPDATE 2

Feature-complete beta by Nov 24

SPRINT 2 · NOV 10–17

- CameraX photos, SHA-256 at capture
- Placement photo for the sensor
- Notices stored on the server
- Deploy backend, geocode address

SPRINT 3 · NOV 17–24

- On-device PDF evidence packet
- Reminders after certification
- TalkBack and font-size pass
- Device tests in CI

Questions?

github.com/Priyanshusingh09876/Habitable